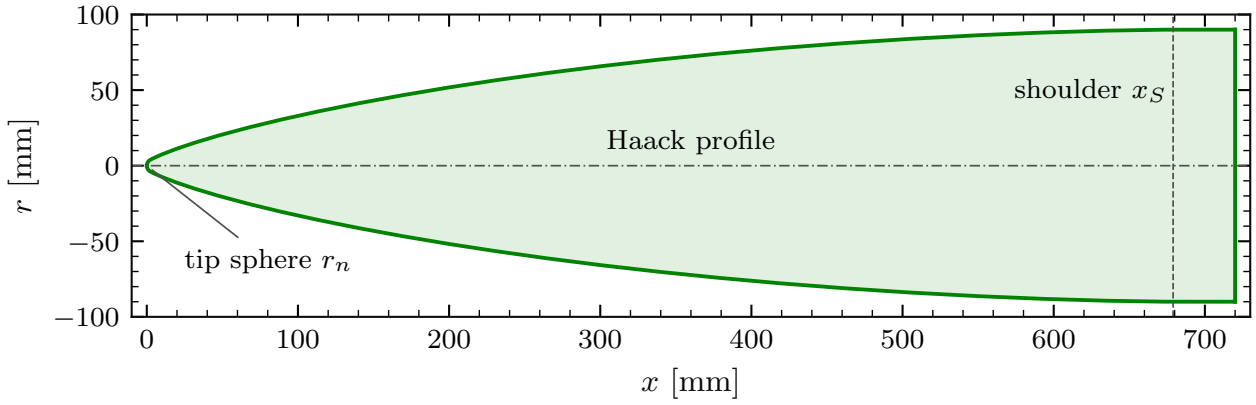
Weisshorn nose cone at 720 mm: geometry for SolidWorks

2027_C_FD_NOSE-CONE-720MM_GEOMETRY — Fabio Feig (FD), 30.10.2026

Design A: LV-Haack, 3U payload inside the nose (baseline)

LV-Haack profile, $679.0 + 41 = 720.0$ mm. x along the axis from the blunted tip, r the outer radius, mm.



Spherical tip $0 \le x \le x_T$: $r = \sqrt{r_n^2 - (r_n - x)^2}$

Haack profile $x_T \le x \le x_S$: $r = \dfrac{R}{\sqrt{\pi}}\sqrt{\theta - \dfrac{\sin 2\theta}{2} + C\sin^3\theta}, \qquad x = \dfrac{L}{2}(1 - \cos\theta) - \delta, \quad \theta_T \le \theta \le \pi$

Coupler cylinder $x_S \le x \le x_S + c$: $r = R$

Parameter	Value
Haack constant C	1/3
Length to the sharp apex L	682.345
Body radius R	90.000 ($D = 180$)
Tip radius r_n	4.500
Length removed by the tip δ	3.345
Tangency point (x_T, r_T)	(2.434, 3.998)
Tangency angle θ_T [rad]	0.1843209
Shoulder x_S	679.000
Coupler cylinder c	41.000
Total length	**720.000**

Tangent to the sphere at x_T and to the cylinder at x_S ($\mathrm{d}r/\mathrm{d}x = 0$).

x	r	x	r	x	r
0.000	0.000	250.000	59.269	525.000	85.042
2.434	3.998	275.000	62.625	550.000	86.307
25.000	13.022	300.000	65.750	575.000	87.404
50.000	20.651	325.000	68.657	600.000	88.333
75.000	27.183	350.000	71.356	625.000	89.087
100.000	33.006	375.000	73.857	650.000	89.652
125.000	38.295	400.000	76.167	675.000	89.983
150.000	43.149	425.000	78.293	679.000	90.000
175.000	47.635	450.000	80.239	700.000	90.000
200.000	51.796	475.000	82.010	720.000	90.000
225.000	55.666	500.000	83.610		

SolidWorks

Sketch on the Front Plane, origin at the blunted tip, x (rocket axis) to the right, document units mm.

1. **Tip:** Centerpoint Arc, centre (4.500, 0), from (0, 0) to (x_T, r_T) = (2.434, 3.998).
2. **Haack profile:** Tools → Sketch Entities → Equation Driven Curve → *Parametric*; paste (angles in radians, t is θ):

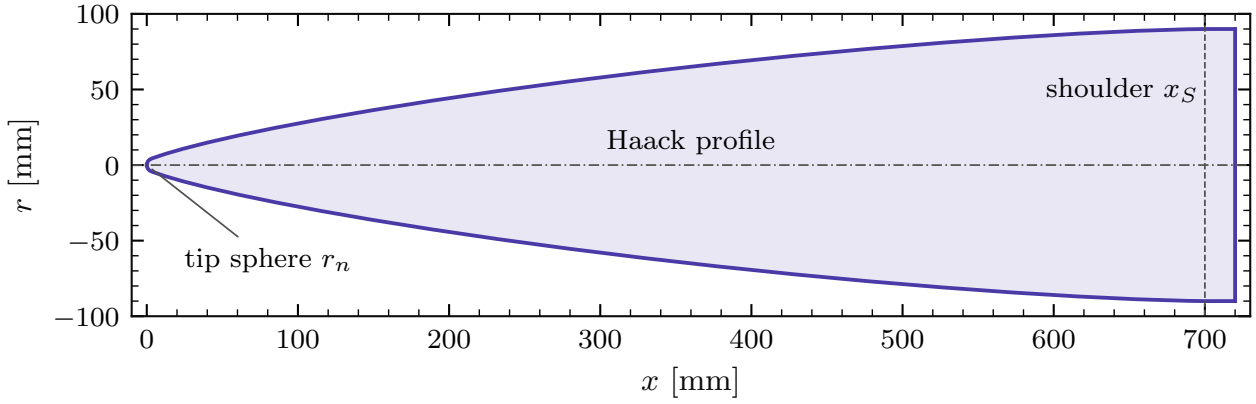
```
x_t = 341.172513*(1 - cos(t)) - 3.345026
y_t = 50.777063*sqr(t - sin(2*t)/2 + 0.3333333*sin(t)^3)
t1  = 0.1843209
t2  = pi
```

   The curve starts on the arc end (x_T, r_T) and ends at (x_S, R) = (679.000, 90.000); add a Tangent relation between the arc and the curve.
3. **Coupler:** Line from (679.000, 90.000) to (720.000, 90.000), then a line down to the axis and a centreline along the axis from the tip.
4. **Solid:** Revolved Boss, 360° about the centreline (outer mould line; shell and joint by ST).

Without equation-driven curves: Insert → Curve → Curve Through XYZ Points with the point file `_solidworks_xyz.txt` (tip to shoulder, X along the axis, $Y = r$, $Z = 0$), then steps 3–4.
Files: `nose_lvhaack_L682_c41_rn4.5mm` + `.step` (solid) and `_solidworks_xyz.txt` (points).

Design B: Von Kármán, 3U payload not in the nose (backup)

Von Kármán profile, $700.0 + 20 = 720.0$ mm. x along the axis from the blunted tip, r the outer radius, mm.



Spherical tip $0 \le x \le x_T$: $r = \sqrt{r_n^2 - (r_n - x)^2}$

Haack profile $x_T \le x \le x_S$: $r = \dfrac{R}{\sqrt{\pi}}\sqrt{\theta - \dfrac{\sin 2\theta}{2} + C\sin^3\theta}, \qquad x = \dfrac{L}{2}(1 - \cos\theta) - \delta, \quad \theta_T \le \theta \le \pi$

Coupler cylinder $x_S \le x \le x_S + c$: $r = R$

Parameter	Value
Haack constant C	0
Length to the sharp apex L	705.465
Body radius R	90.000 ($D = 180$)
Tip radius r_n	4.500
Length removed by the tip δ	5.465
Tangency point (x_T, r_T)	(2.916, 4.212)
Tangency angle θ_T [rad]	0.2184214
Shoulder x_S	700.000
Coupler cylinder c	20.000
Total length	**720.000**

Tangent to the sphere at x_T and to the cylinder at x_S ($\mathrm{d}r/\mathrm{d}x = 0$).

x	r	x	r	x	r
0.000	0.000	250.000	51.442	525.000	80.831
2.916	4.212	275.000	54.781	550.000	82.712
25.000	11.036	300.000	57.975	575.000	84.444
50.000	17.201	325.000	61.030	600.000	86.013
75.000	22.609	350.000	63.953	625.000	87.400
100.000	27.535	375.000	66.746	650.000	88.578
125.000	32.106	400.000	69.412	675.000	89.494
150.000	36.393	425.000	71.952	700.000	90.000
175.000	40.443	450.000	74.366	720.000	90.000
200.000	44.287	475.000	76.653		
225.000	47.948	500.000	78.809		

SolidWorks

Sketch on the Front Plane, origin at the blunted tip, x (rocket axis) to the right, document units mm.

1. **Tip:** Centerpoint Arc, centre (4.500, 0), from (0, 0) to $(x_T, r_T) = (2.916, 4.212)$.
2. **Haack profile:** Tools → Sketch Entities → Equation Driven Curve → *Parametric*; paste (angles in radians, t is θ):

```
x_t = 352.732280*(1 - cos(t)) - 5.464559
y_t = 50.777063*sqr(t - sin(2*t)/2)
t1  = 0.2184214
t2  = pi
```

   The curve starts on the arc end (x_T, r_T) and ends at $(x_S, R) = (700.000, 90.000)$; add a Tangent relation between the arc and the curve.
3. **Coupler:** Line from (700.000, 90.000) to (720.000, 90.000), then a line down to the axis and a centreline along the axis from the tip.
4. **Solid:** Revolved Boss, 360° about the centreline (outer mould line; shell and joint by ST).

Without equation-driven curves: Insert → Curve → Curve Through XYZ Points with the point file `_solidworks_xyz.txt` (tip to shoulder, X along the axis, $Y = r$, $Z = 0$), then steps 3–4.
Files: `nose_vonkarman_L705_c20_rn4.5mm` + `.step` (solid) and `_solidworks_xyz.txt` (points).